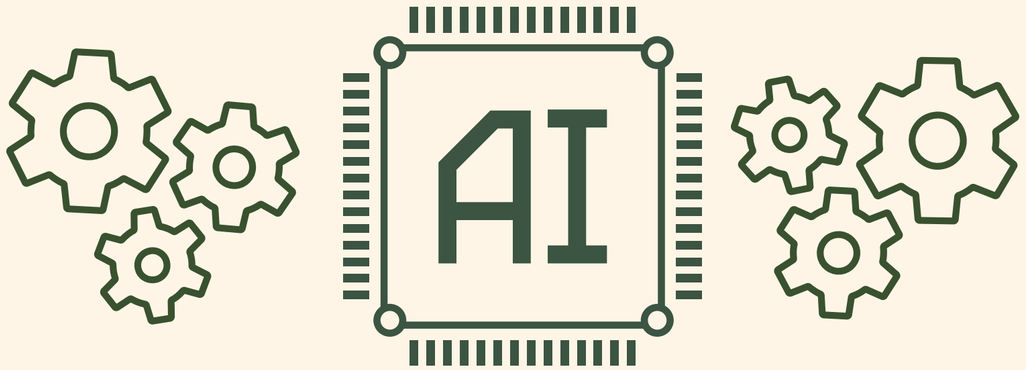
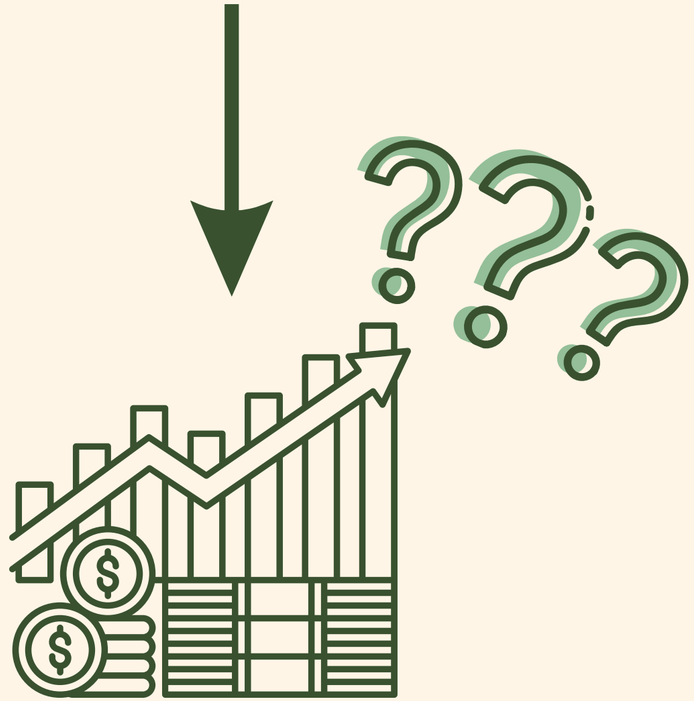
AI Productivity: Does AI Actually Make Companies More Money ?



A data analysis of 3,248 tasks across teams, task types and pricing models.



MACHINE LEARNING – A.A. 2025/26

MALTIDE MARUCCI • CAMILLA CORTOPASSI • FRANCESCO COCCHI

AGENDA

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OUR QUESTIONS AND PROMPTS

THE AI PRODUCTIVITY PARADOX

Companies using AI get things done faster. But their profits don't always go up.

Billable Hours Drop

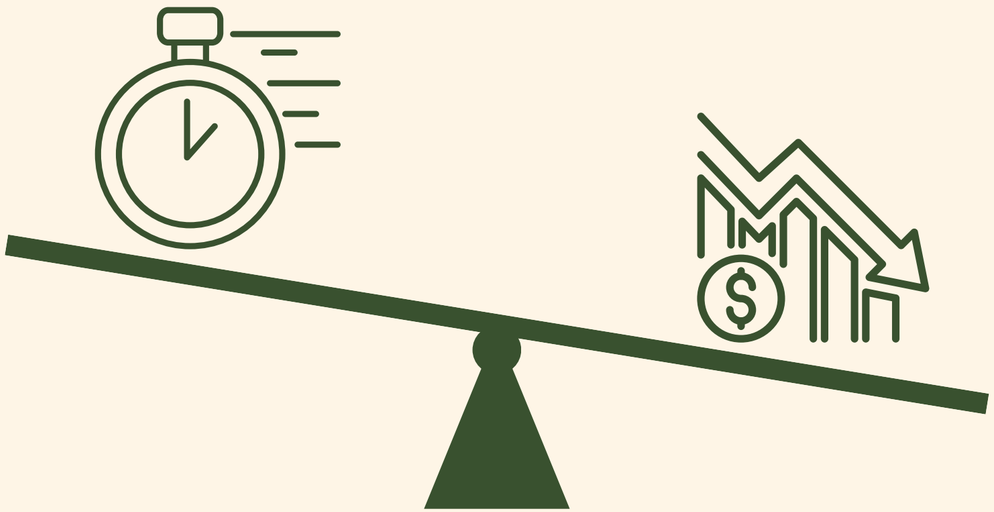
AI cuts hours worked → fewer billable hours in hourly pricing models

Partial Adoption Instability

Partial adoption creates instability and unexpected rework

Pricing Models Lag Behind

Pricing models haven’t caught up with how AI changes the work



HOW WE DID IT

The Dataset

3,248 tasks with hours, rework, quality scores, revenue, cost and AI usage: a comprehensive view of real work output.

How we manage the dataset - pipeline

Clean data **Handle missing** **Explore patterns** **Build features** **Run tests**

Four Core Questions

- Where is value created?
- Where do losses come from?
- Quality vs. speed trade-off?
- What is the critical adoption threshold?

Three Advanced Questions

- Speed vs Quality Trade-off
- Rework Threshold
- AI - > Real Profit ?

3 KEY INSIGHTS

1

More AI = Better Margins But Only Past 30%

AI-assisted tasks double the profit margin (0.16 vs 0.08), but only when rework costs are excluded. Once factored in, the 75–100% group is the only one staying positive across all scenarios. The 15–30% band is the danger zone.

2

AI Makes Delivery Faster, Not Better

AI cuts delivery time by ~13 hours per task (4.43 vs 4.96 days) and reduces SLA breaches from 47% to 38%, both significant at all levels. Output quality, however, shows no measurable difference ($p = 0.60$, 68.86 vs 69.21)

3

Senior Workers Lose Money Under Hourly Pricing

Seniors generate negative margins across all AI usage levels, the hourly model fails to recover their cost regardless of adoption. Juniors and mids stay profitable. The root cause: seniors use AI less frequently and less effectively, creating a double problem of pricing and AI adoption.

3 KEY INSIGHTS *FIRST INSIGHT*

T-TEST FOR VALUE CREATION

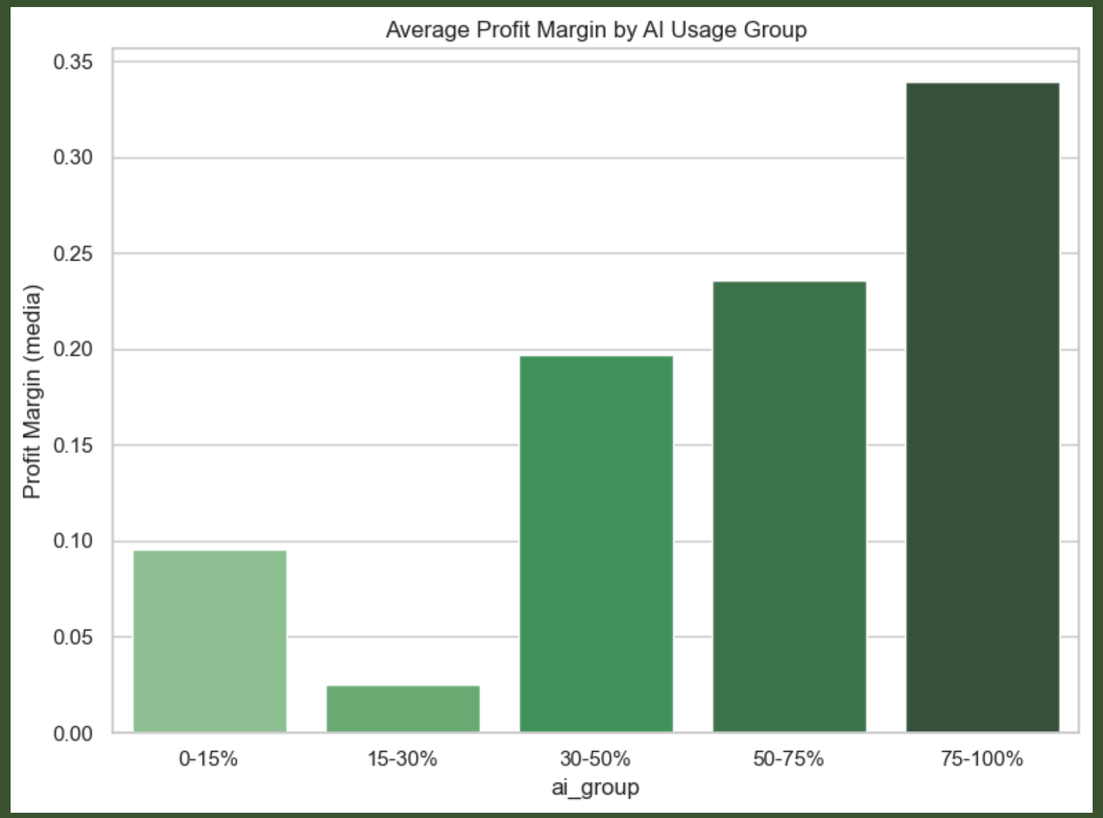
To compare AI-assisted and non-assisted tasks we use a two sample t-test, which tests whether the difference in group means is likely due to chance or reflects a real underlying pattern.
We report results at three significance levels (10%, 5%, 1%) to give a more complete picture rather than a binary yes/no.

PROFIT MARGIN

AI assisted: 0.16 | No AI: 0.08

t-test - p=0.0698
α=0.10 → Significant
α=0.05 → Not significant
α=0.01 → Not significant

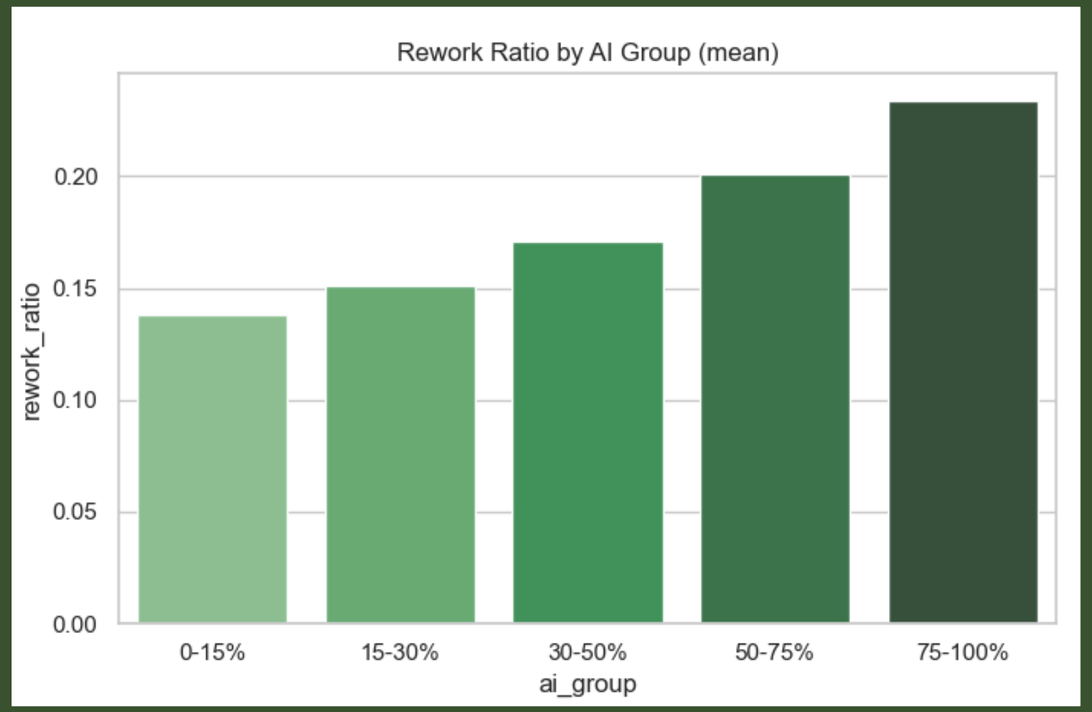
Looking at financial performance, AI-assisted tasks show a higher average profit margin (0.16 vs 0.08) significant at the 10% level.



3 KEY INSIGHTS *FIRST INSIGHT*

While the previous slide showed that higher AI usage tends to be associated with higher profit margins, rework costs were not taken into account.

As visible here, the rework ratio consistently increases with AI usage, suggesting that higher AI adoption also brings more corrective work.



REWORK_RATIO = REWORK_HOURS / (HOURS_SPENT + REWORK_HOURS)

Since the true cost of rework hours is uncertain, we modeled three scenarios:

1. **Optimistic**: rework hours billed at 50% of the standard hourly cost, assuming rework is simpler and requires fewer resources
2. **Normal (baseline)**: rework hours billed at the standard hourly cost (1×)
3. **Pessimistic**: rework hours billed at 150% of the standard hourly cost, reflecting higher complexity or senior resource involvement

These scenarios allow us to reassess profitability by incorporating rework costs, and to evaluate how sensitive the results are to different assumptions about their true cost.

3 KEY INSIGHTS *FIRST INSIGHT*

To assess the true financial impact of AI adoption, profit margins were recalculated for each scenario by adding rework costs back into the cost base: the higher the rework cost assumption, the greater the erosion of margin.

Since higher AI usage is associated with more rework hours, one would expect higher AI groups to suffer greater margin erosion once rework costs are factored in. However, the results reveal a more nuanced picture:

The **75-100% group** is the only one maintaining **positive** or **near-zero margins** across all three scenarios, confirming that very high AI adoption remains resilient even after accounting for rework costs.
The **50-75% group** shows the most severe sensitivity, reaching approximately **-85%** margin in the **Pessimistic scenario**, suggesting this bucket carries high rework exposure without the full efficiency gains of the top group.

3 KEY INSIGHTS *SECOND INSIGHT*

To test whether AI has an impact on delivery speed we use a two-sample t-test for delivery time and a chi-square test for SLA breach, which is the appropriate choice for binary variables.

Delivery time is highly significant ($p < 0.01$), AI-assisted tasks are delivered faster on average: 4.43 days (~106 hours) vs 4.96 days (~119 hours), a difference of about 13 hours per task.

SLA breach tells the same story: AI-assisted tasks breach the SLA 38% of the time compared to 47% for non-assisted ones, a difference significant at all three levels. Speed gains appear to be real and statistically robust.

On the **quality side**, a t-test on the outcome score shows no statistically significant difference between AI-assisted and non-assisted tasks ($p \sim 0.60$). The values are virtually identical (68.86 vs 69.21) a gap of less than 0.4 points on a 0–100 scale, suggesting that AI usage has no measurable impact on output quality, at least as captured by outcome score alone.

DELIVERY TIME

AI assisted: 4.43 | No AI: 4.91

t-test p=0.0000
α=0.10 -> Significant
α=0.05 -> Significant
α=0.01 -> Significant

SLA BREACH

AI assisted: 38.1% | No AI: 47.4%

chi-square p=0.0006
α=0.10 -> Significant
α=0.05 -> Significant
α=0.01 -> Significant

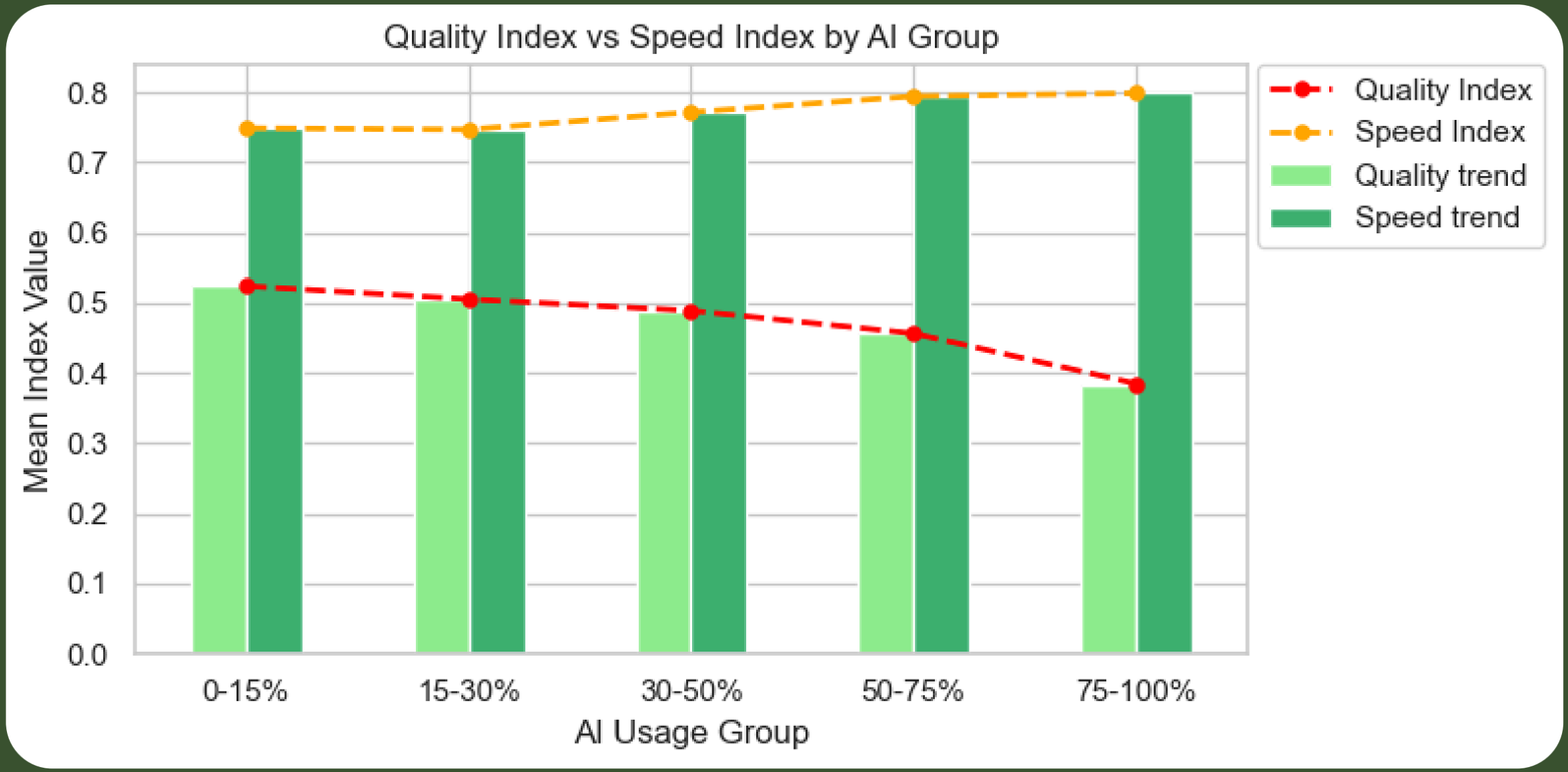
OUTCOME SCORE

AI assisted: 68.86 | No AI: 69.21

t-test p=0.5969
α=0.10 -> Not significant
α=0.05 -> Not significant
α=0.01 -> Not significant

3 KEY INSIGHTS *SECOND INSIGHT*

To get a broader picture, we also built a Quality Index (normalized outcome score minus rework ratio) and a Speed Index (based on how far delivery is from the SLA).



The Speed Index improves with AI usage, though not consistently, confirming the t-test results on delivery time.

SPEED_INDEX = 1 - SLA_RATIO_NORM

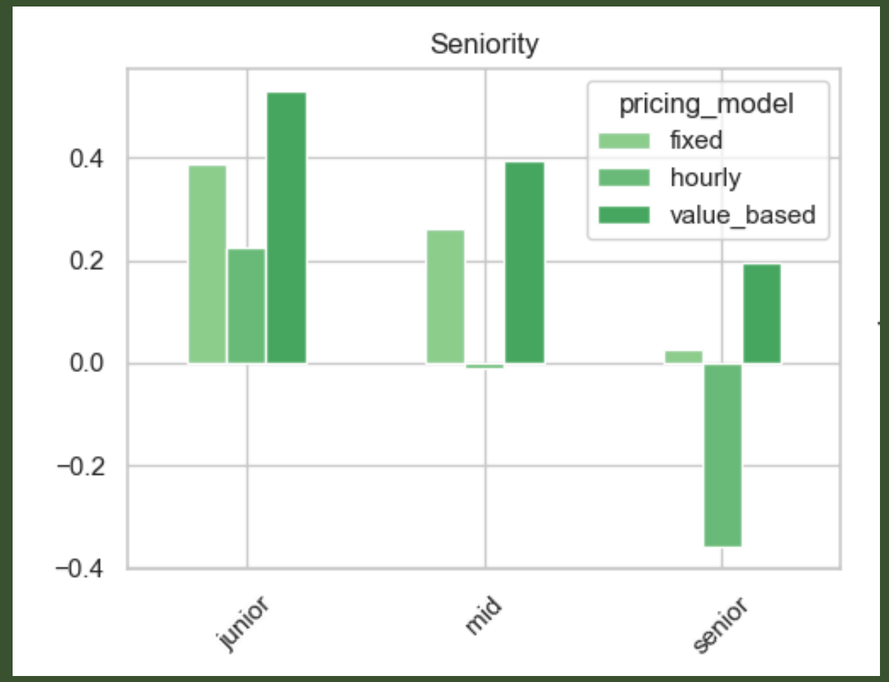
The Quality Index instead declines steadily, from 0.52 in the 0–15% group down to 0.38 in the 75–100% group.

QUALITY_INDEX = OUTCOME_NORM - REWORK_NORM

3 KEY INSIGHTS *THIRD INSIGHT*

To understand when the hourly pricing model becomes unsustainable, we compared it against fixed and value-based pricing across six potential loss drivers: rework, errors, complexity, hours spent, billable hours and seniority.

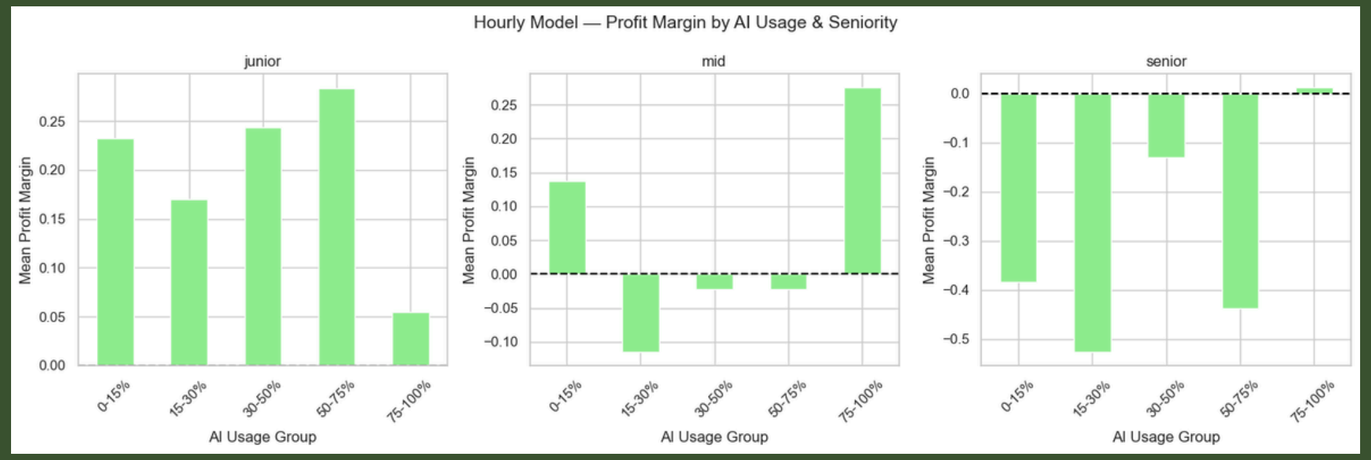
Only seniority showed a clear signal.
Under the hourly pricing model, senior professionals consistently generate negative margins, while junior and mid-level profiles remain profitable.



This raises two critical questions:

1. Is this a people problem or an AI problem?
2. Does seniority determine not just whether AI is used, but how effectively?

3 KEY INSIGHTS *THIRD INSIGHT*



Seniors generate near-zero or negative margins across all AI usage levels, the hourly model fails to recover their cost regardless of AI adoption.

Juniors remain profitable across most buckets, with highest adopters performing best, AI is a genuine productivity

Mid-level profiles show instability, recovering only at the highest AI usage levels.

Why? Junior profiles, more naturally exposed to AI, adopt it more effectively. Seniors, despite domain expertise, use these tools less frequently and less effectively, limiting their ability to offset their higher cost.

Our answers:
seniors are less trained to leverage AI effectively, and the hourly model amplifies the damage:
a double problem, of pricing and of how they use AI.

CONCRETE DECISION

Our recommendation comes down to three concrete actions:

Set a minimum AI usage threshold. The data shows that partial adoption (between 20% and 30%) is the riskiest configuration. Teams should either commit to AI or not use it at all. We recommend setting a minimum threshold of 30% AI usage for any AI-assisted engagement.

Reprice senior profiles on hourly contracts. Senior professionals generate negative margins under the hourly model regardless of how much AI they use. The fix is not to push them to use more AI , it is to move them to a fixed or value-based pricing model where their expertise, not their hours, determines the price.

Invest in AI training for senior profiles. The gap between junior and senior performance is not just a pricing issue: it is an adoption issue. Juniors leverage AI naturally and effectively; seniors do not. Structured AI training for senior professionals could help close this gap and partially offset their higher cost base.

BEHIND THE NUMBERS *1 THINGS DISCOVERED THANKS TO AI*

We learned **threshold analysis**.

Working with continuous variables like AI usage, the natural approach is to look at averages, but averages hide turning points.

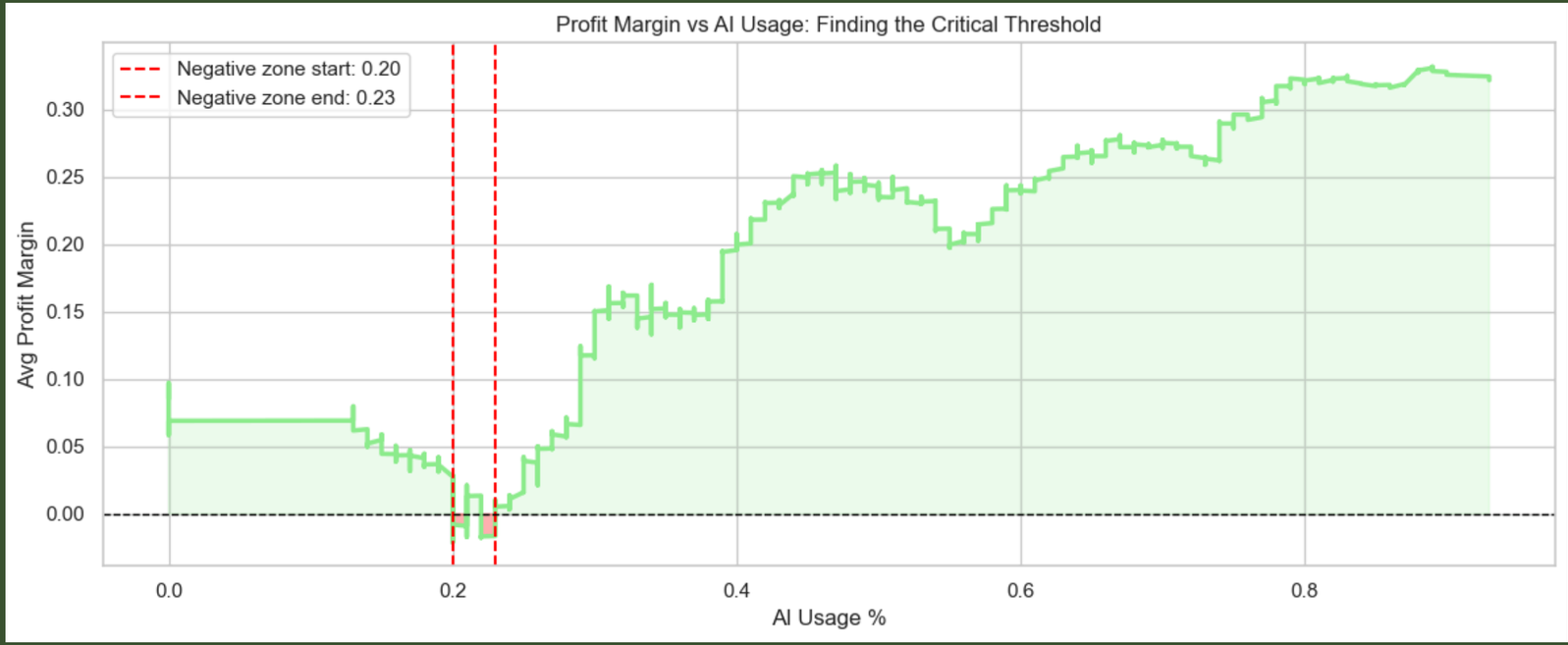
→

AI helped us go further: by applying a **rolling average** over the 500 nearest observations, we identified exactly when AI usage starts hurting margins, without any statistical assumption.

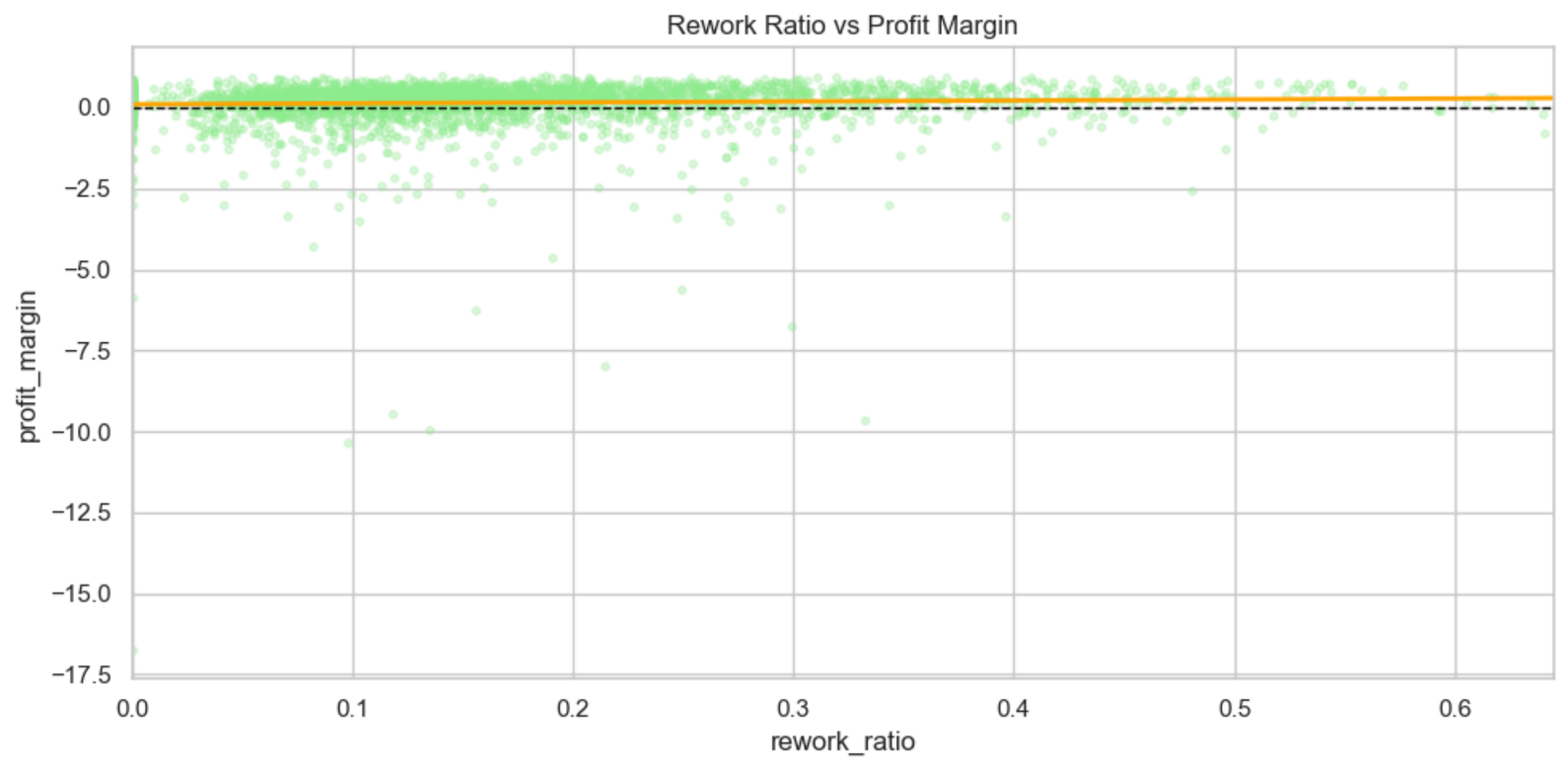
The result:

- **Below 20%** = stable margin (~6%)
- **20–25%** = margin turns negative -> critical threshold
- **Above 30%** = steady recovery and growth (~30%)

Partial adoption is the riskiest configuration and we would not have found this without AI-assisted exploration.



BEHIND THE NUMBERS *1 MISTAKES MAKE BY AI*



AI Misread the Rework Signal

Early in the analysis, AI-assisted interpretation suggested that high rework ratio was a reliable predictor of margin erosion . Intuitively, **more rework = more cost = lower profit**.

Correlation: 0.0519, p = 0.0031

Pearson r ≈ 0.008 (p =) 0.638

Rework ratio has almcost no correlation with profit margin.

The Real Driver

The actual driver is cost ratio and pricing model, not how many times a task was revised. Rework is a symptom, not the cause.

OUR QUESTIONS AND PROMPTS *ONE EXAMPLE*

EXAMPLES OF QUESTIONS THAT EMERGED DURING THE WORKFLOW:

Does AI usage reduce the share of loss-making tasks, and by how much compared to low-adoption groups?
Does higher AI usage reduce cost ratio, or does it add overhead that offsets the efficiency gains?

Example: PROMPTS FOR VALUE METRICS:

1

I have a dataframe with several performance metrics and a bool column ai_assisted. I want to statistically test whether using AI has a significant effect on those metrics. What's a good approach?

2

Ok great so t-test makes sense. My metrics are profit_margin, profit_per_hour, cost_ratio, efficiency, revenue. Can you show me how to split the final_df by ai_assisted and run a ttest_ind for each one?

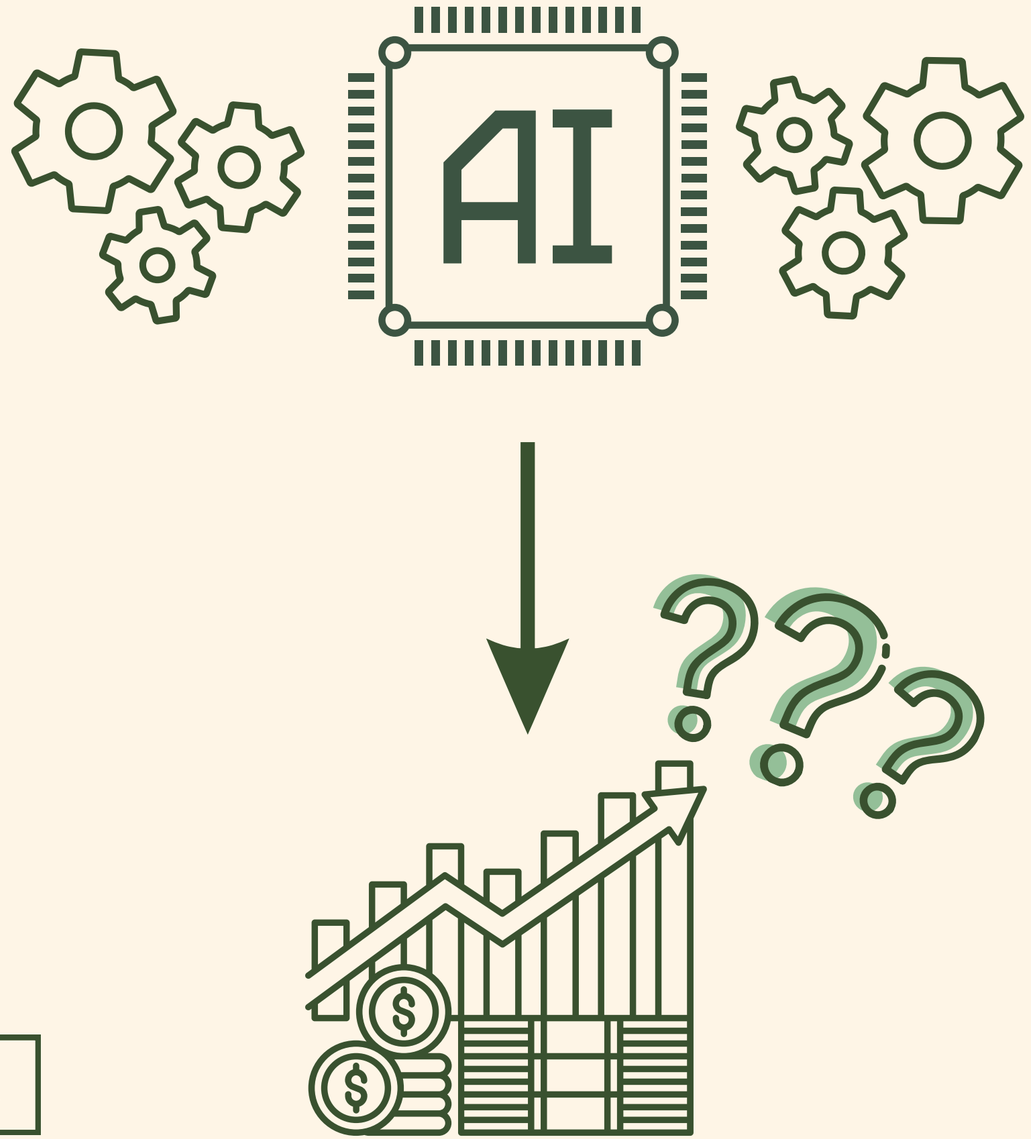
3

The output only prints p-value but I want to also see the values for both groups side by side, so I can actually tell if AI is better or worse, not just whether it's significant

4

You used only alfa = 0.01 but I want to show all three thresholds: 0.10, 0.05 and 0.01, since at 0.01 none of the results are significant. A result significant at 0.10 or 0.05 but not 0.01 is still useful information for me, I don't want to throw it away.

THANK YOU
for the attention !



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